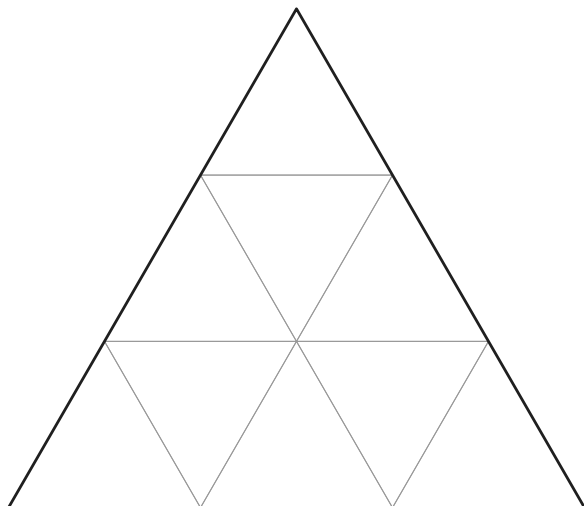
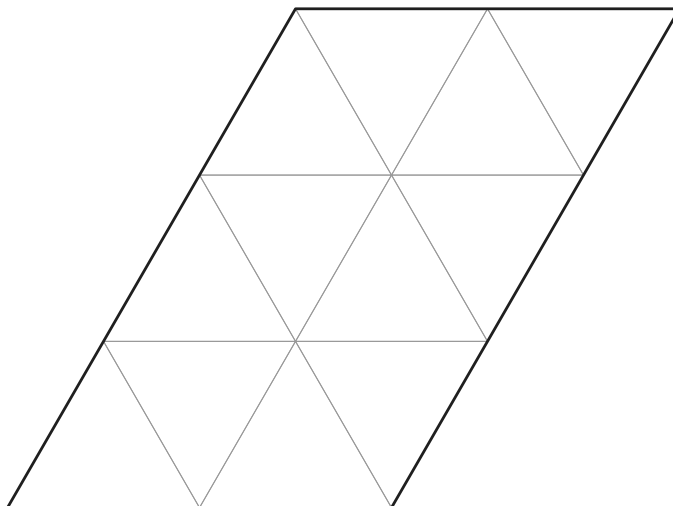
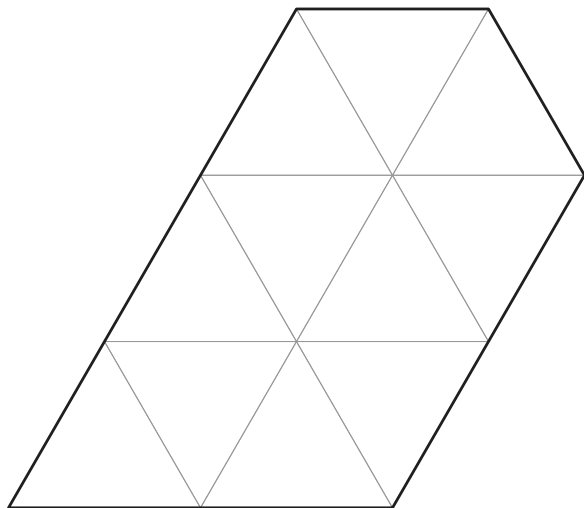
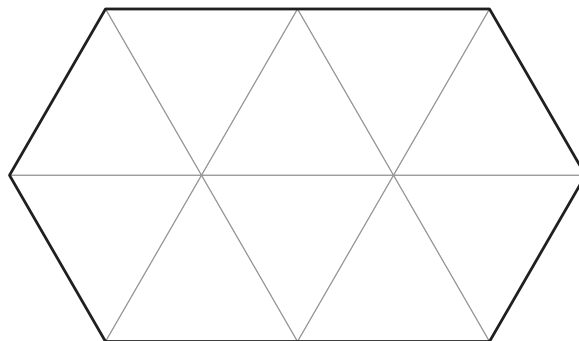


Version W

Week 1 tiling · Grades 2–3

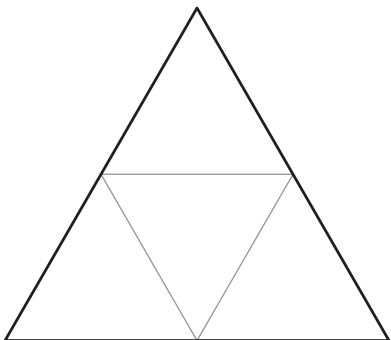
Problem 1: Try to fill each board using only small blue rhombi. Each blue covers two whole triangle spaces. Keep pieces inside the board without overlaps.

Board A**Board B****Board C****Board D**

Which boards can you fill and which ones did you get stuck on? What is stopping you from completely filling some of the shapes?

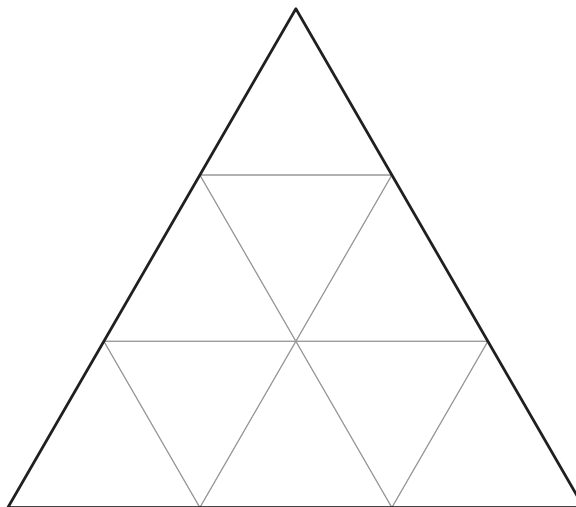
Problem 2: Use only small blue rhombi on these triangles and the triangle on the next page. Fit as many as you can, without overlaps or pieces outside. Record how many blues fit and how many triangle spaces stay empty.

2 small edges on each side



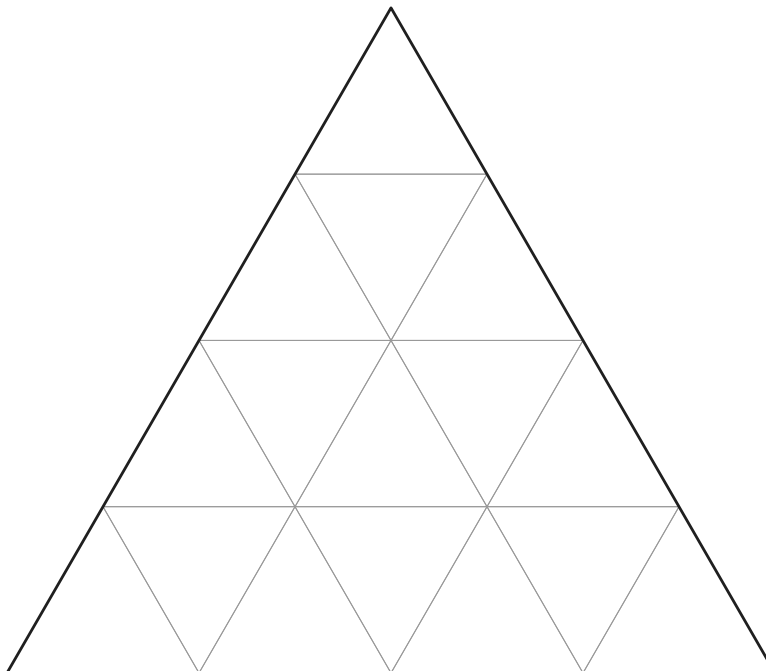
Blues: _____ Empty spaces: _____

3 small edges on each side



Blues: _____ Empty spaces: _____

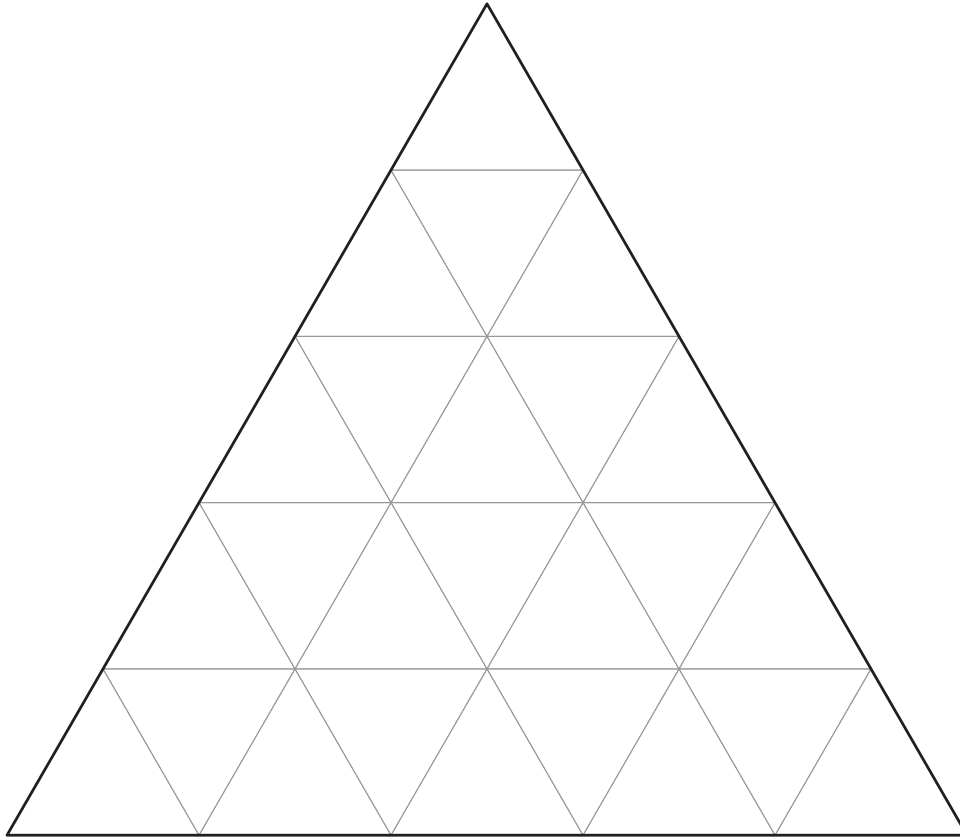
4 small edges on each side



Blues: _____ Empty spaces: _____

Problem 2: Fit as many small blue rhombi as you can in this triangle. Keep pieces inside the triangle without overlaps. Record your best packing.

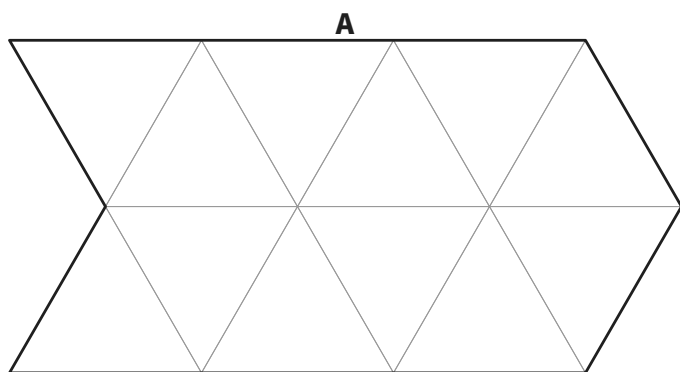
5 small edges on each side



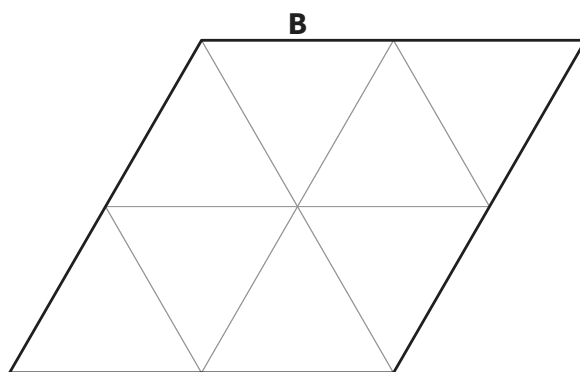
Blues: _____ Empty spaces: _____

Compare your four best packings. What do you notice about the number of empty spaces? Explain why you cannot leave fewer.

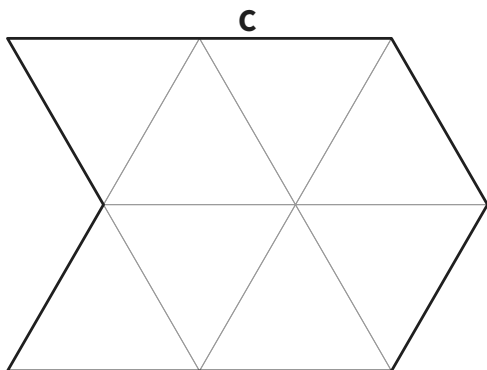
Problem 3: On each board, first use only small blue rhombi. Then clear it and use only purple chevrons. Cover whole triangle spaces without overlaps or pieces outside. Leave as few spaces empty as you can. Record your best results.



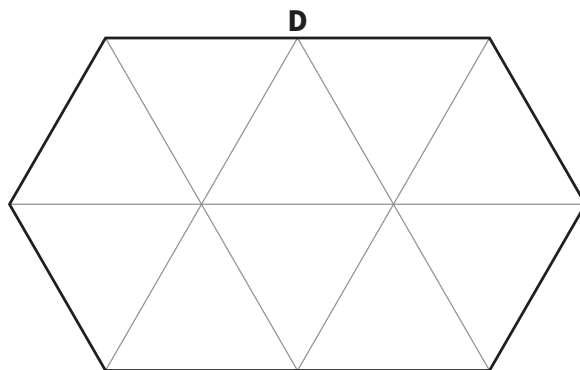
Fewest empty triangle spaces:
Blues: ____ Purples: ____



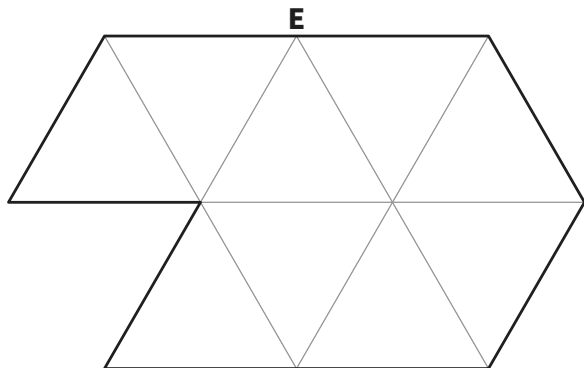
Fewest empty triangle spaces:
Blues: ____ Purples: ____



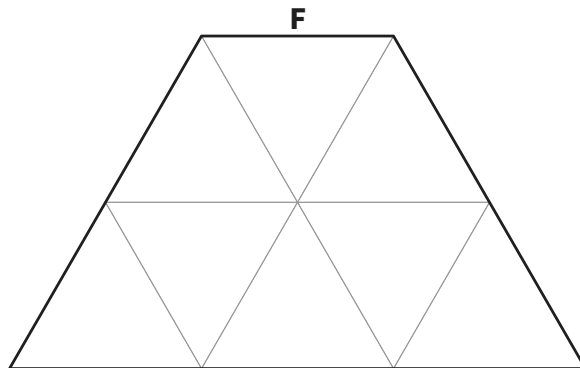
Fewest empty triangle spaces:
Blues: ____ Purples: ____



Fewest empty triangle spaces:
Blues: ____ Purples: ____



Fewest empty triangle spaces:
Blues: ____ Purples: ____



Fewest empty triangle spaces:
Blues: ____ Purples: ____

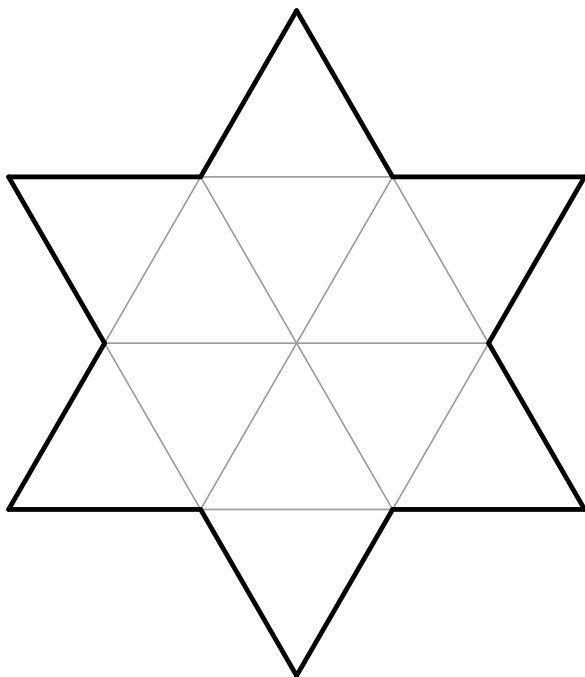
Can you cover one purple exactly with blues? How does this explain which pieces can leave fewer gaps?

Version X

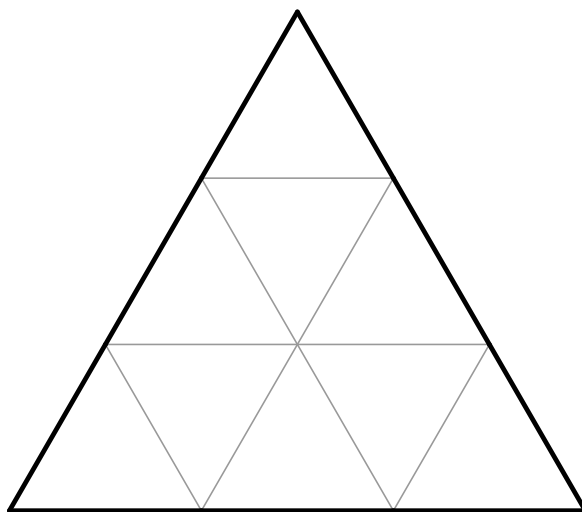
Week 1 tiling · Grades 2–3

On every board in this packet, each block must cover whole small triangles of the board. Blocks may not overlap or stick out past the thick outline.

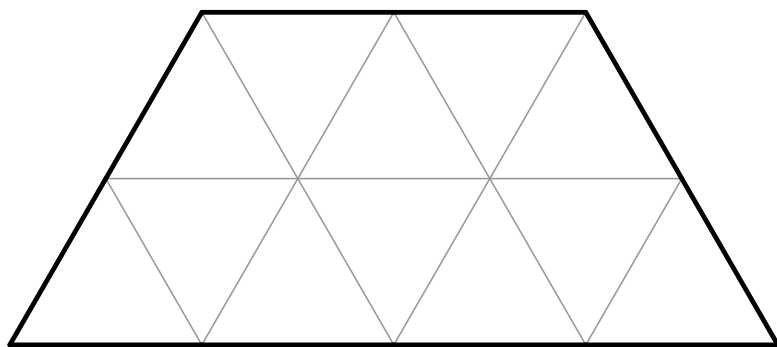
Problem 1: Cover each board with blue rhombuses , or decide that it cannot be done. For each board that cannot be covered, explain why.



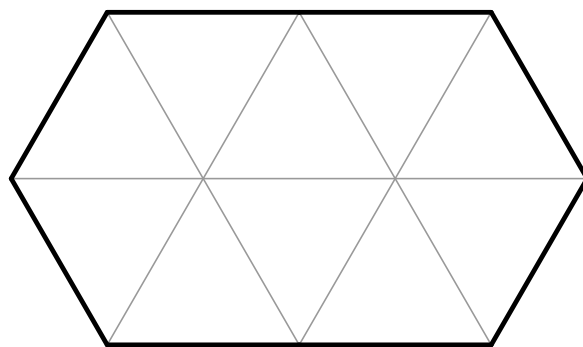
A



B

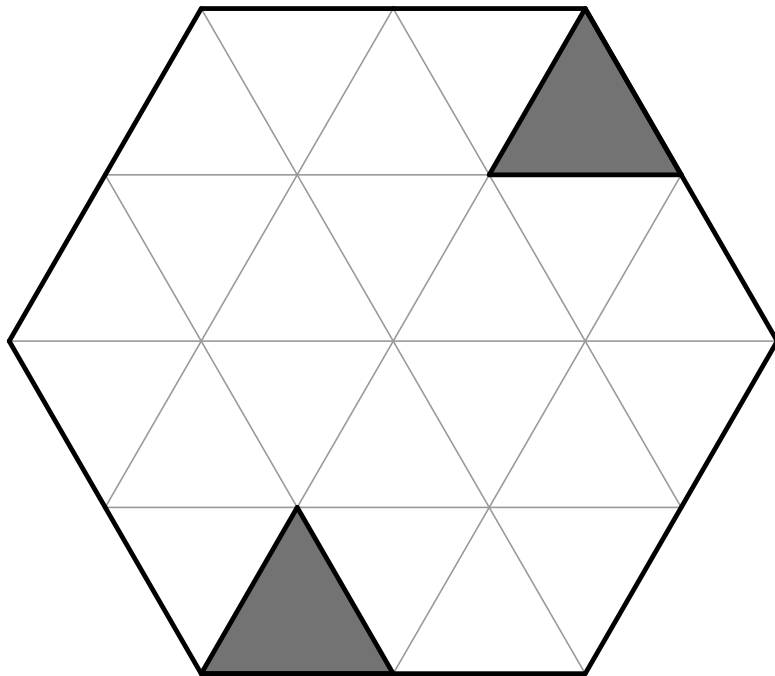


C

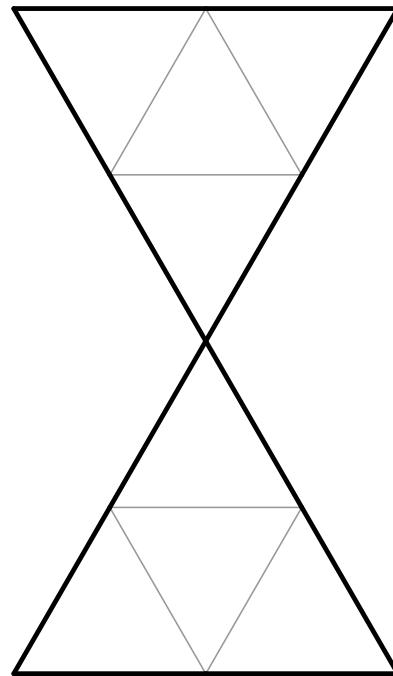


D

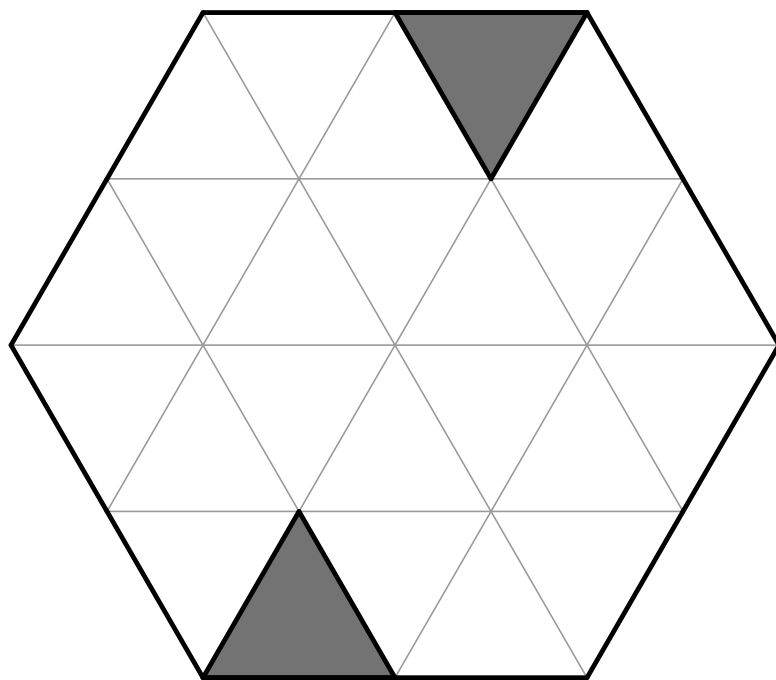
Problem 2: The gray triangles are holes, so they are not part of the board. Cover each board with blue rhombuses , or decide that it cannot be done. For each board that cannot be covered, explain why.



E

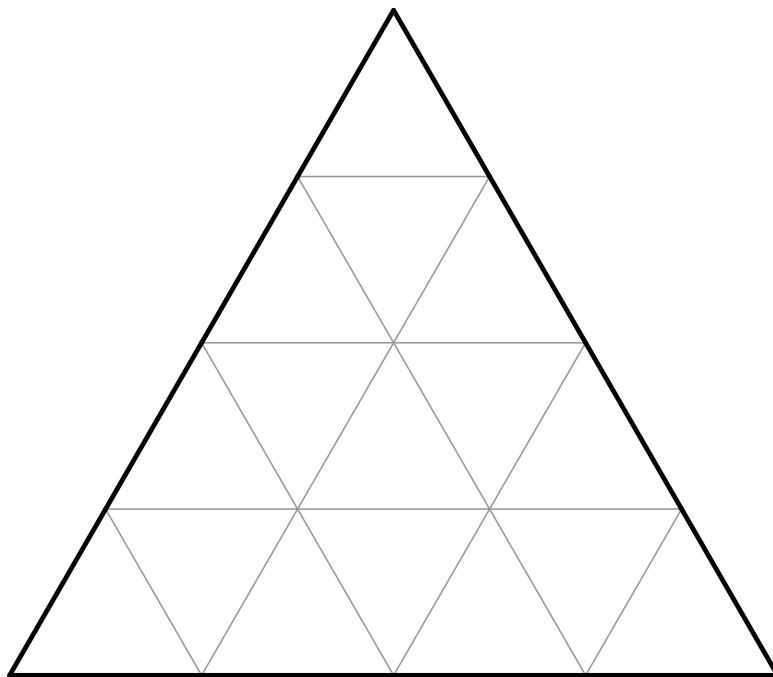
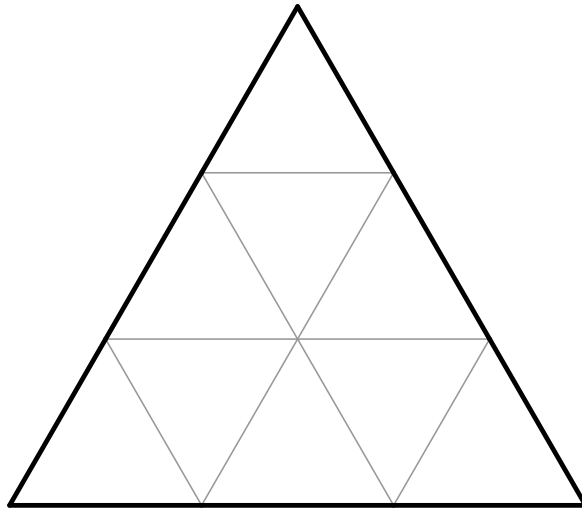
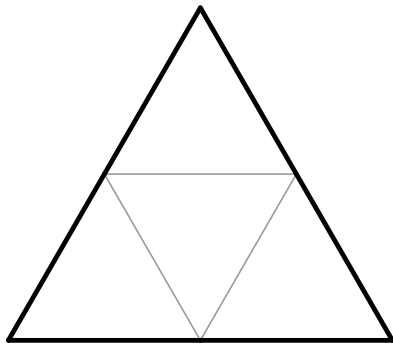


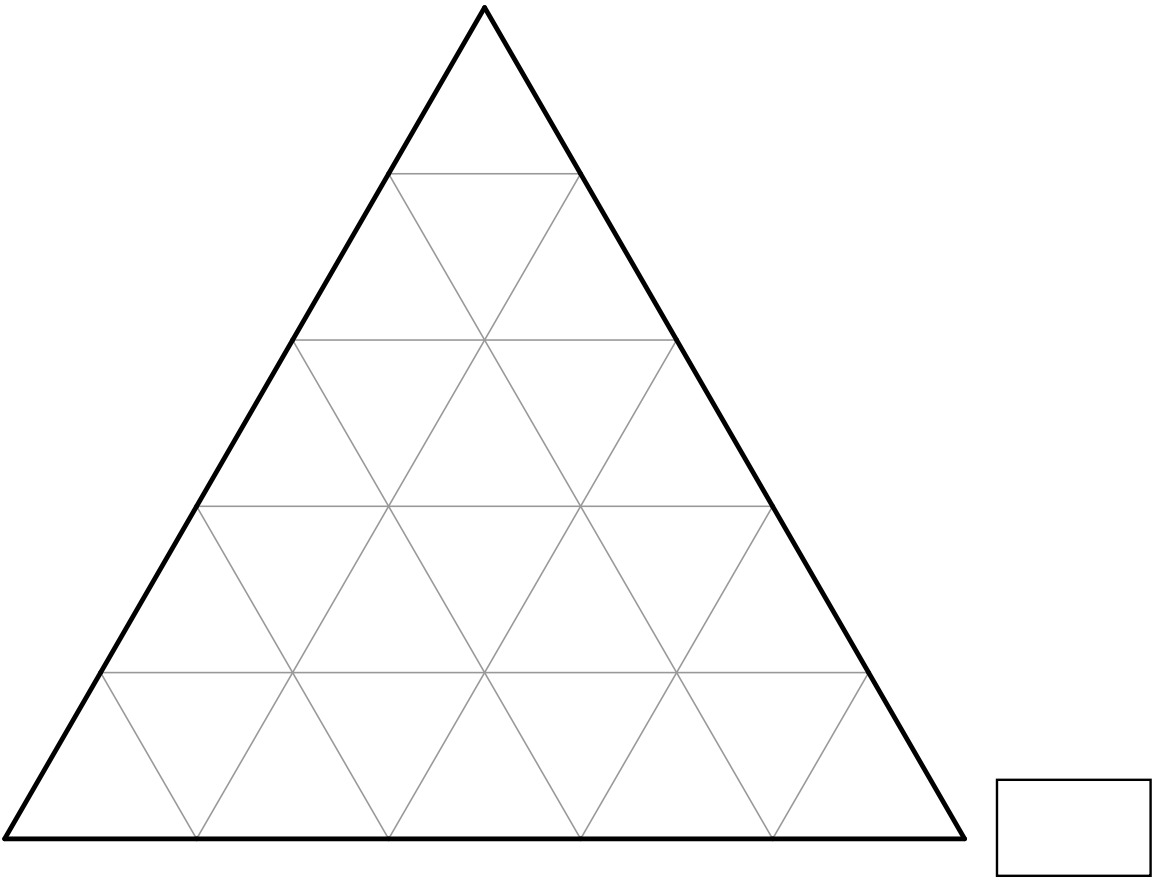
F



G

Problem 3: Cover each big triangle with blue rhombuses  and green triangles . Use as few green triangles as you can, and write that number in the box next to the triangle. Then explain why nobody can cover the biggest triangle with fewer green triangles than you used.



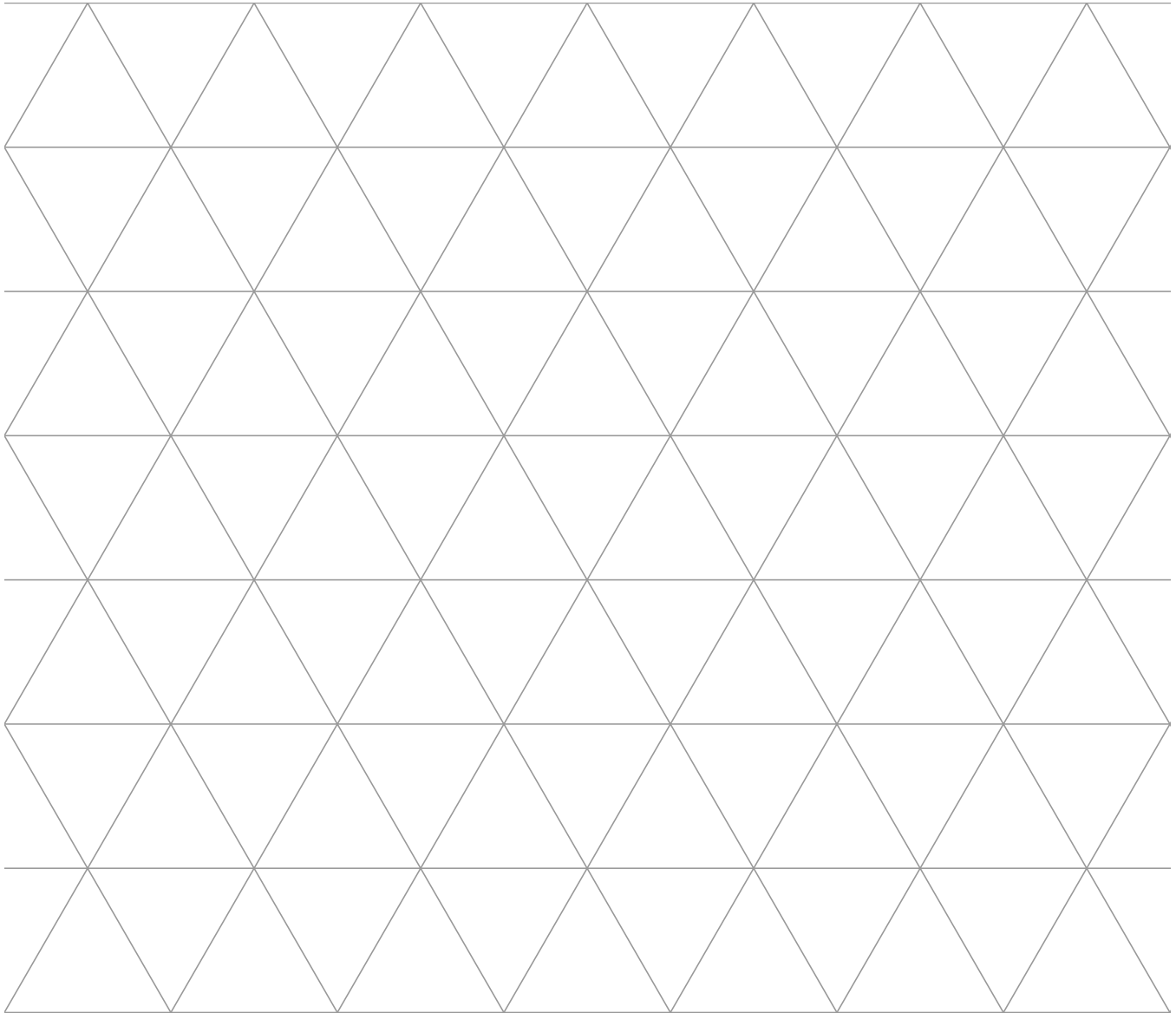


Problem 4: Imagine a triangle board with 10 small triangles along each side. If you cover it with blue rhombuses and green triangles, what is the fewest number of green triangles you need? What if the board has 100 small triangles along each side?

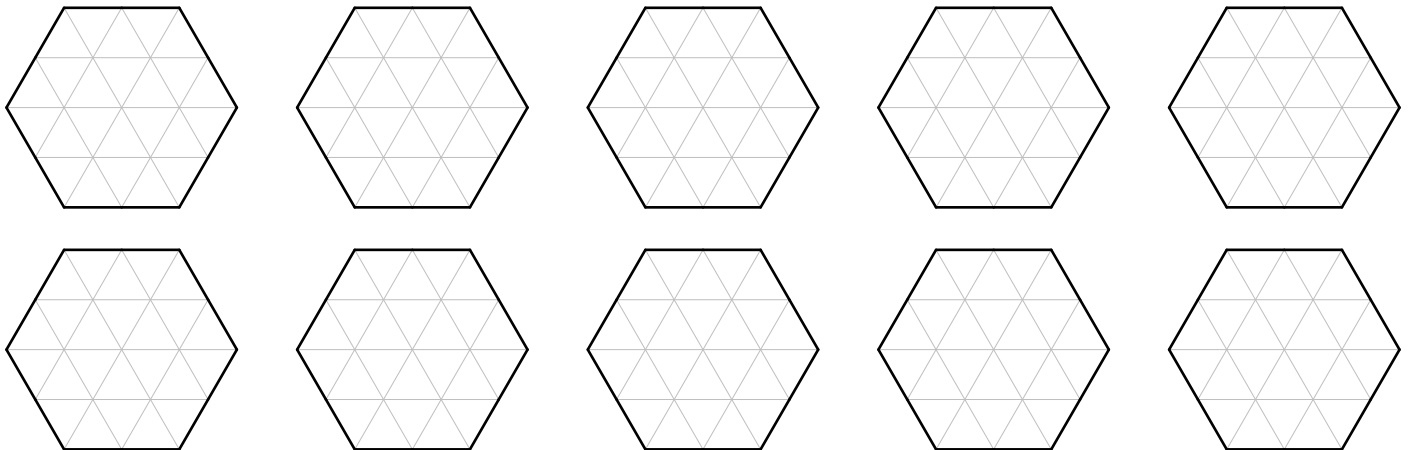
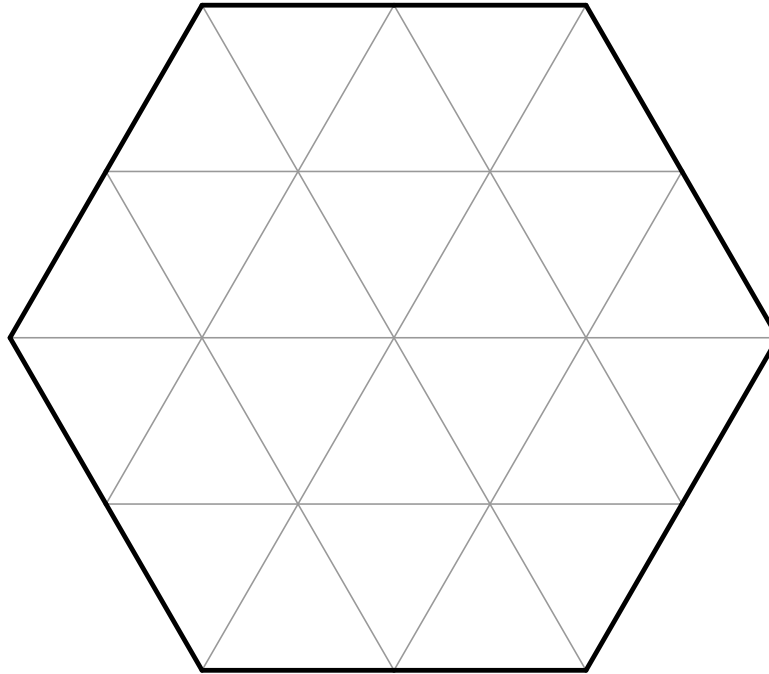
Problem 5: Go back to the four triangles in Problem 3. This time cover each one with purple chevrons  and green triangles  , using as few green triangles as you can. Write your numbers in the table. Can purple chevrons ever need fewer green triangles than blue rhombuses? Explain.

	with blue rhombuses	with purple chevrons
2 small triangles on each side		
3 small triangles on each side		
4 small triangles on each side		
5 small triangles on each side		

Problem 6: On the grid below, draw the outline of a board made of exactly 20 small triangles that needs at least 6 green triangles when you cover it with blue rhombuses and green triangles. The board must stay in one piece if you cut it out.



Problem 7: Put two green triangles  anywhere on this hexagon board. Can blue rhombuses  cover the rest of the board? Find a rule that tells which places for the two green triangles work, and explain why your rule is right. You can mark the places you try on the small hexagons.

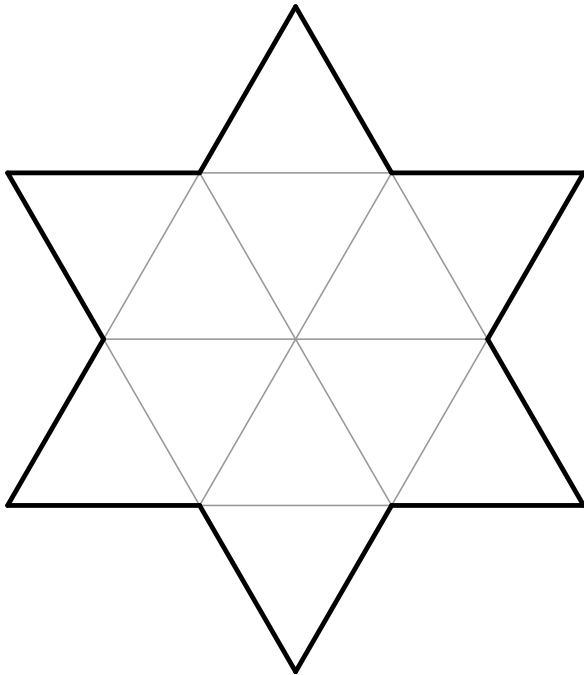


Version Y

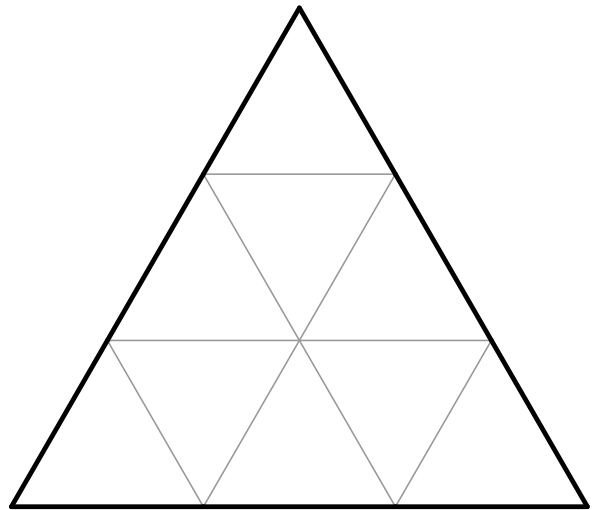
Week 1 tiling · Grades 2–3

Use only the small blocks. On every board in this packet, each block must cover whole small triangles of the board. Blocks may not overlap or stick out past the thick outline.

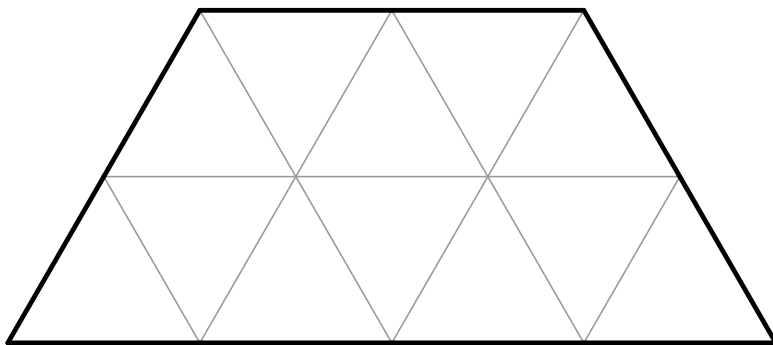
Problem 1: Cover each board with blue rhombuses , or decide that it cannot be done. For each board that cannot be covered, explain why.



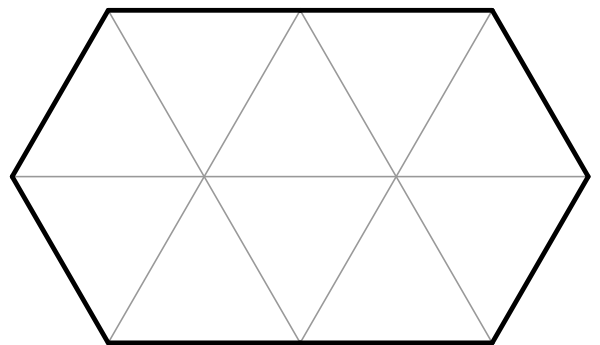
A



B

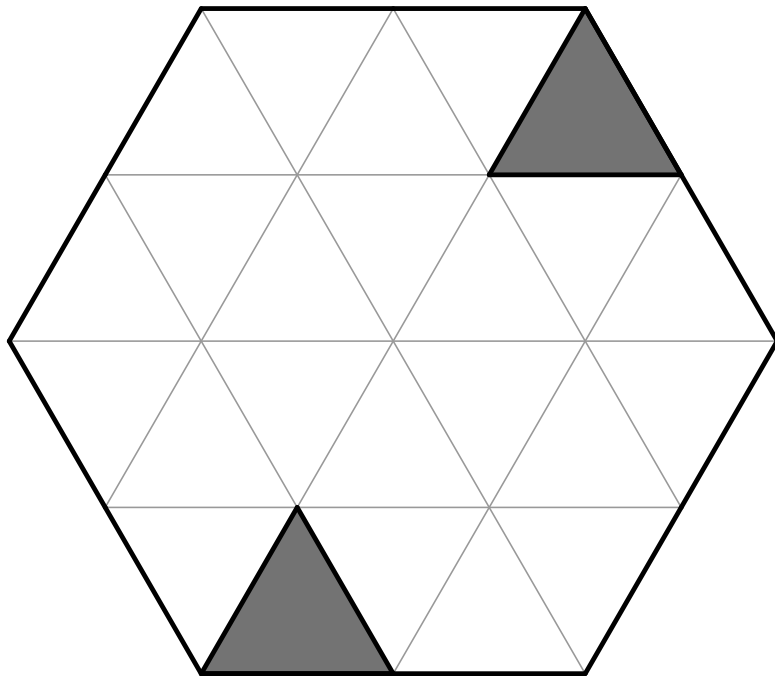


C

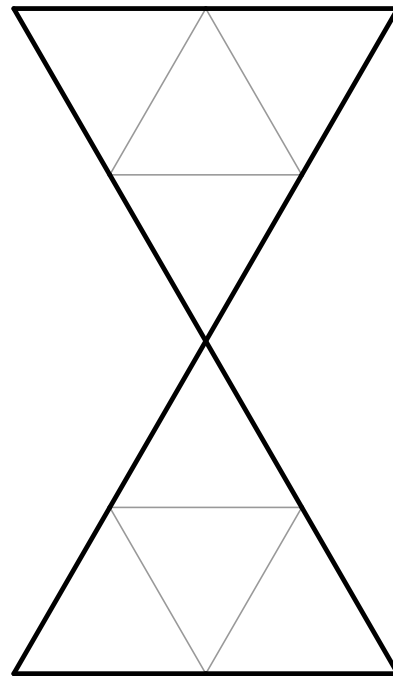


D

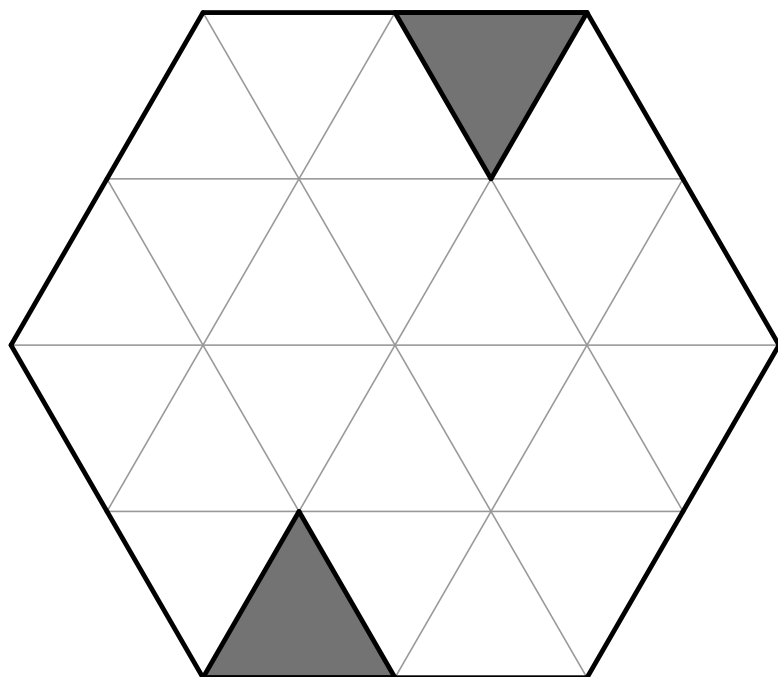
Problem 2: The gray triangles are holes, so they are not part of the board. Cover each board with blue rhombuses , or decide that it cannot be done. For each board that cannot be covered, explain why.



E

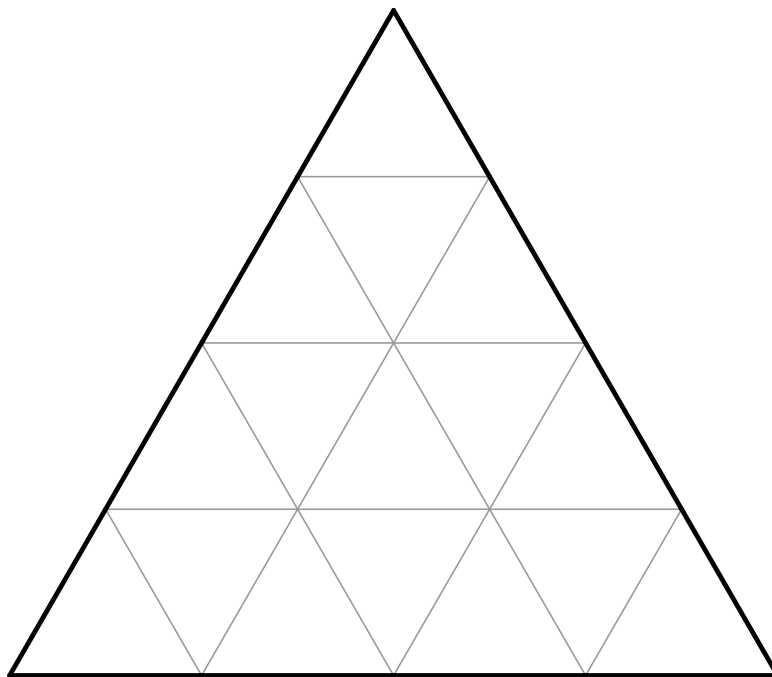
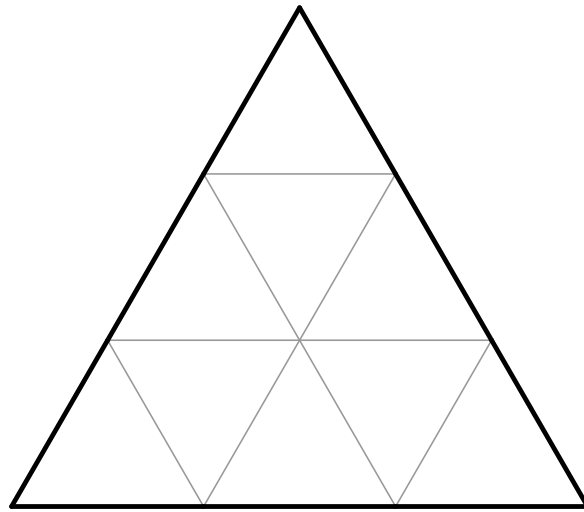
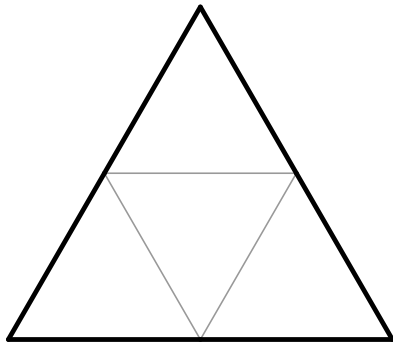


F

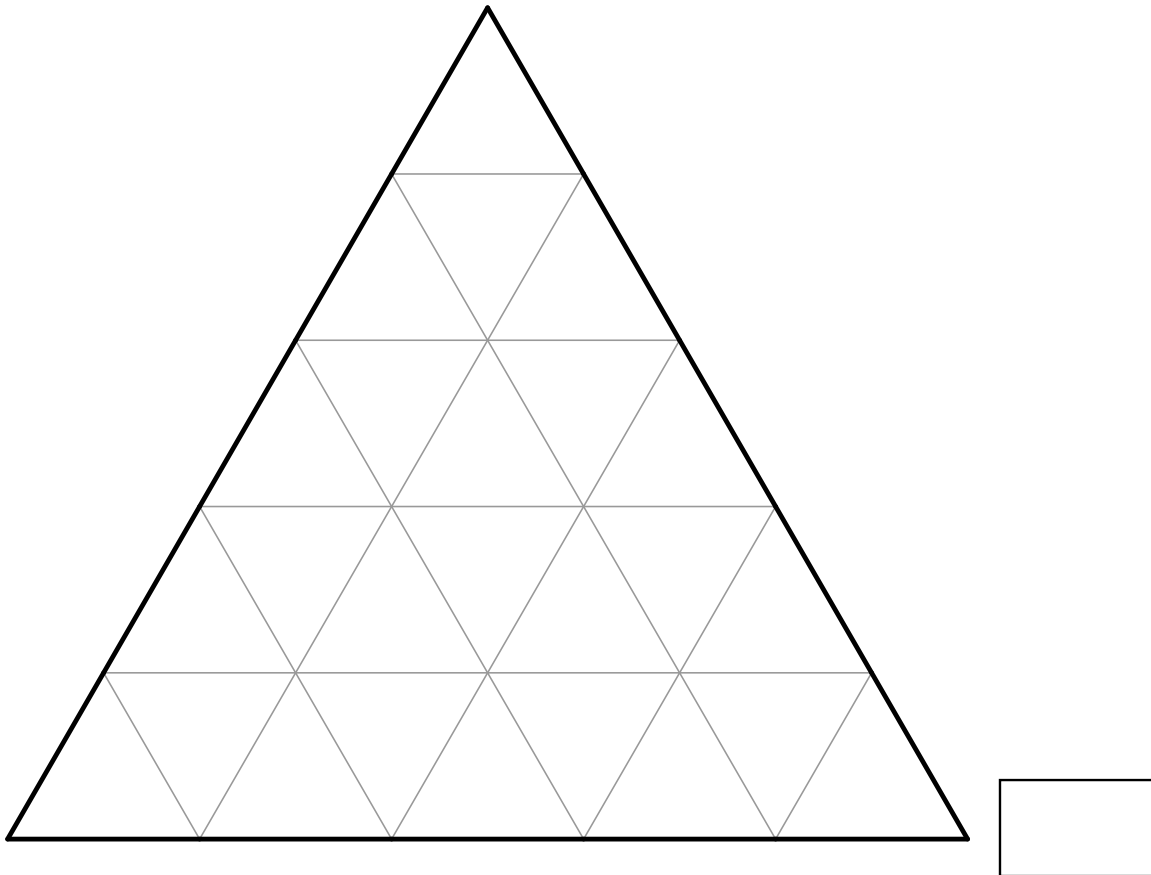


G

Problem 3: Cover each big triangle with blue rhombuses  and green triangles . Use as few green triangles as you can, and write that number in the box next to the triangle.



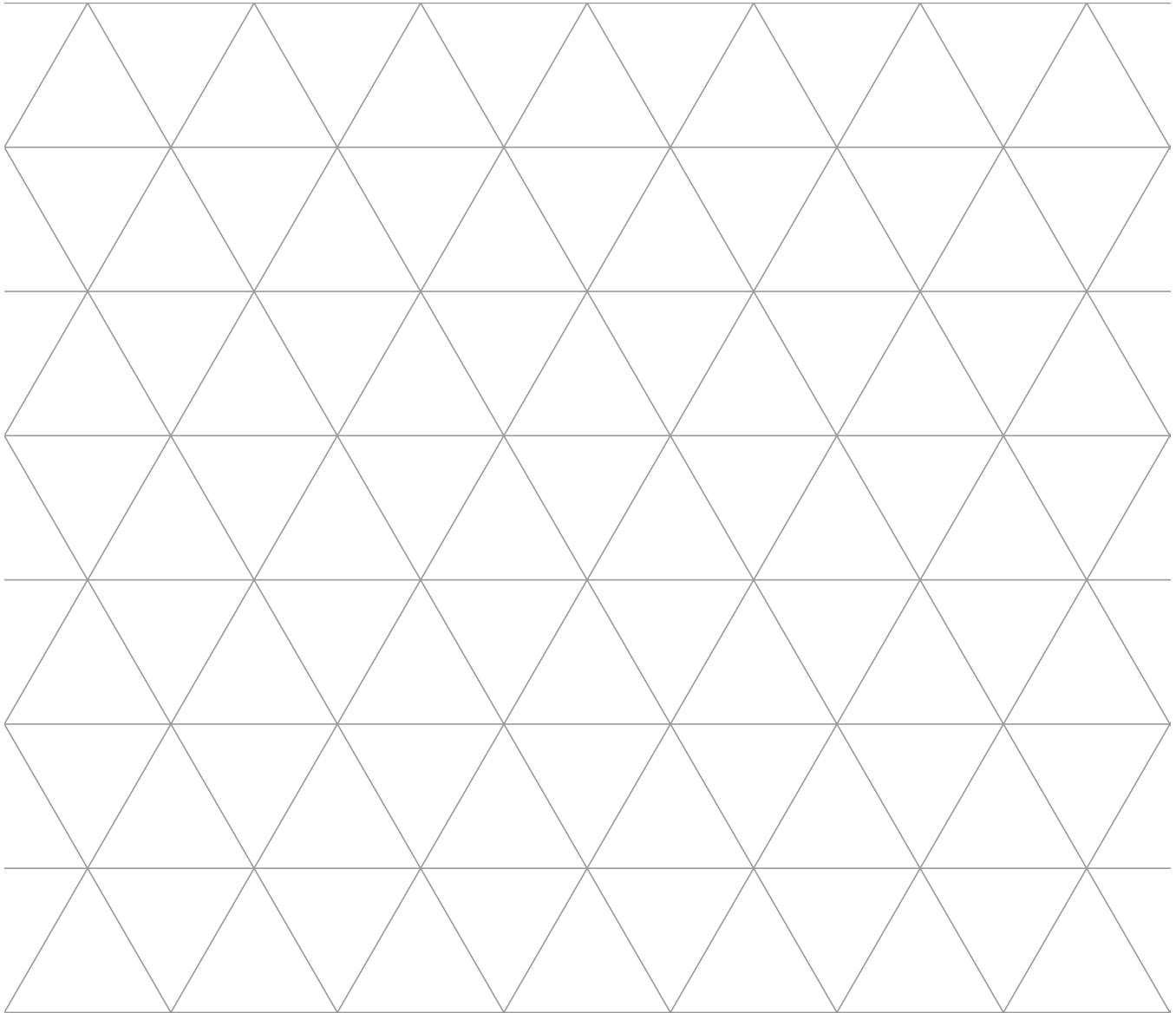
Problem 4: Cover this triangle with blue rhombuses  and green triangles , using as few green triangles as you can, and write that number in the box. How many green triangles would you need for a triangle with 10 small triangles on each side? With 100? Explain why nobody can use fewer green triangles than your answers.



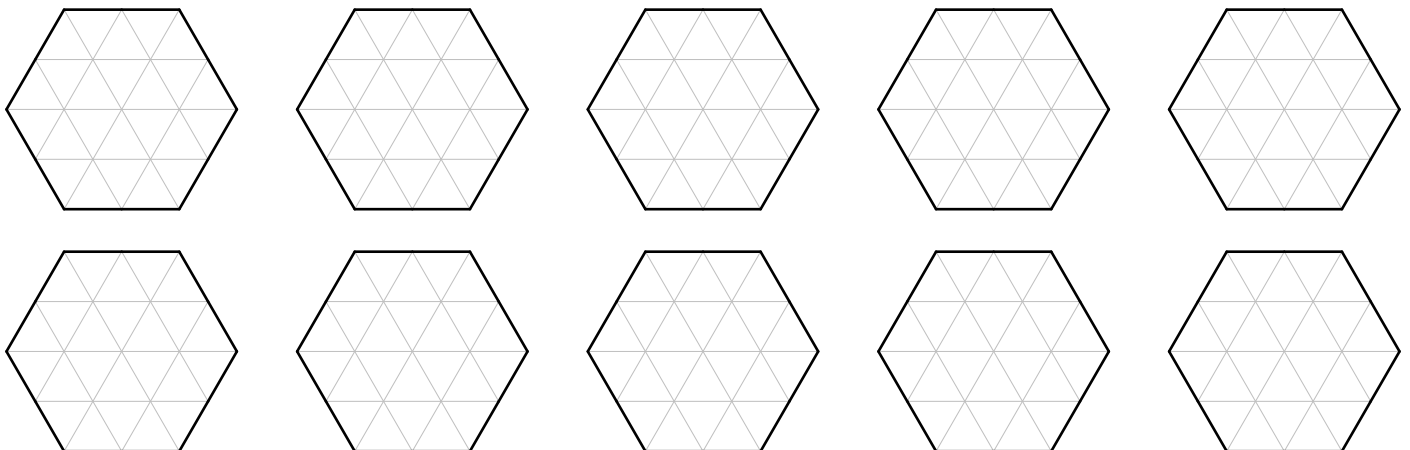
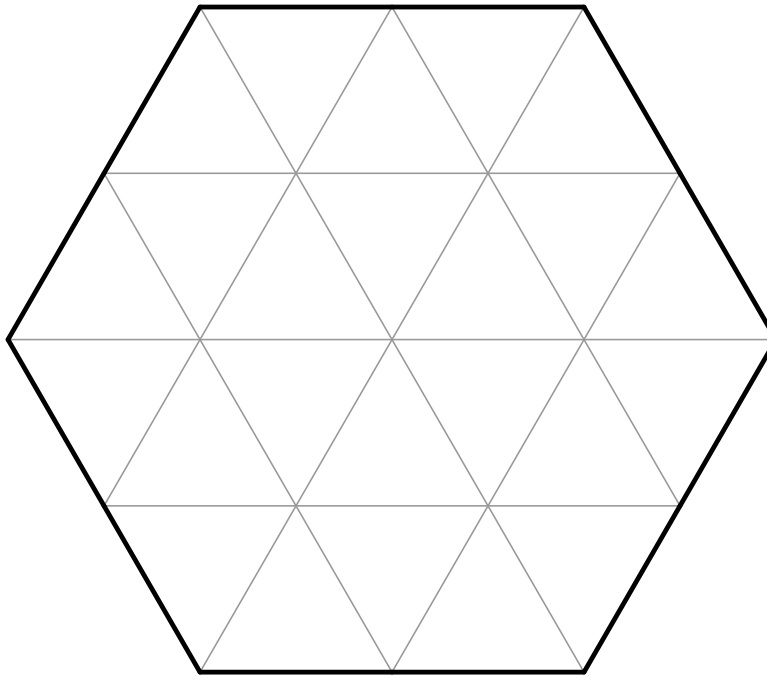
Problem 5: Go back to the four triangles in Problems 3 and 4. This time cover each one with purple chevrons  and green triangles  , using as few green triangles as you can. Write your numbers in the table. Can purple chevrons ever need fewer green triangles than blue rhombuses? Explain.

	with blue rhombuses	with purple chevrons
2 small triangles on each side		
3 small triangles on each side		
4 small triangles on each side		
5 small triangles on each side		

Problem 6: On the grid below, draw the outline of a board made of exactly 20 small triangles that needs at least 6 green triangles when you cover it with blue rhombuses and green triangles. The board must stay in one piece if you cut it out.



Problem 7: Put two green triangles  anywhere on this hexagon board. Can blue rhombuses  cover the rest of the board? Find a rule that tells which places for the two green triangles work. Explain why the places that do not fit your rule can never work. You can mark the places you try on the small hexagons.



Version Z

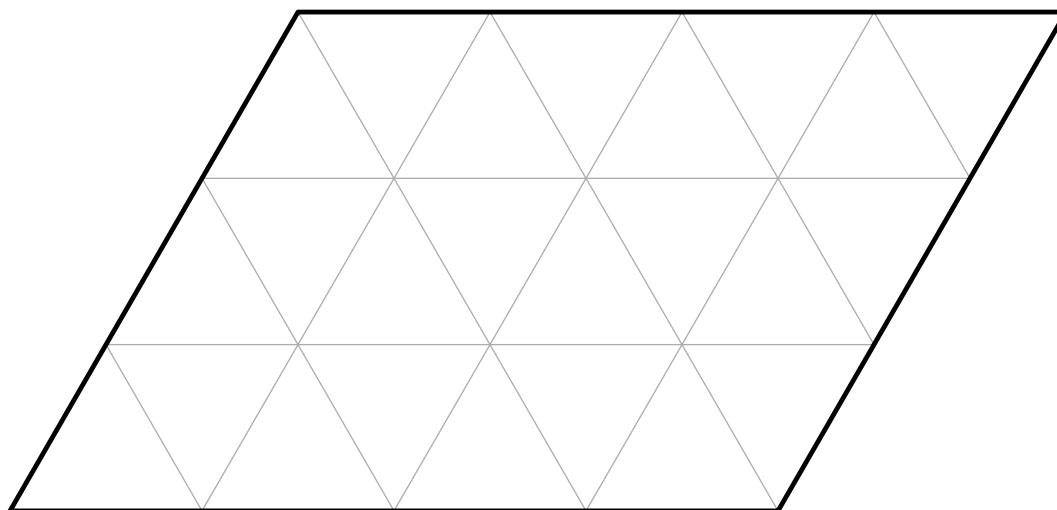
Week 1 tiling · Grades 2–3

1 Ups and downs

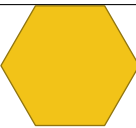
Name: _____

On our boards, every small triangle points **UP** ▲ or **DOWN** ▼.

1. Put a dot in every **UP** ▲ triangle on this board.



2. Put one block at a time on the board, lined up with the lines. How many ▲ and how many ▼ does it cover? Try different spots and turns!

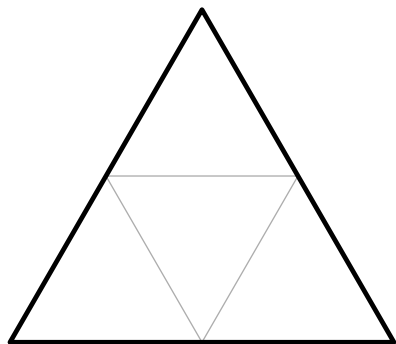
Block	covers ▲	covers ▼
		
		
		
		
		

Talk about it. Did any block give two different answers? Can you place a blue so that it covers two ▲ triangles?

2 Possible or impossible?

Cover a board with **blue rhombi only** : no gaps, no overlaps, nothing hanging over the edge. Circle **Possible** or **Impossible**. Then count the  and  triangles on the board.

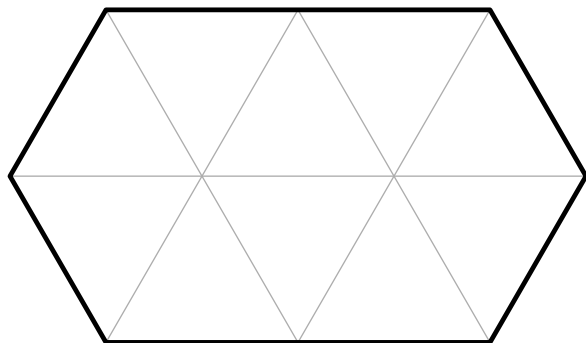
A



Possible Impossible

 _____  _____

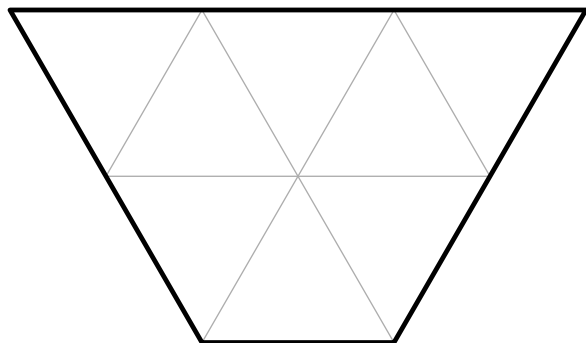
B



Possible Impossible

 _____  _____

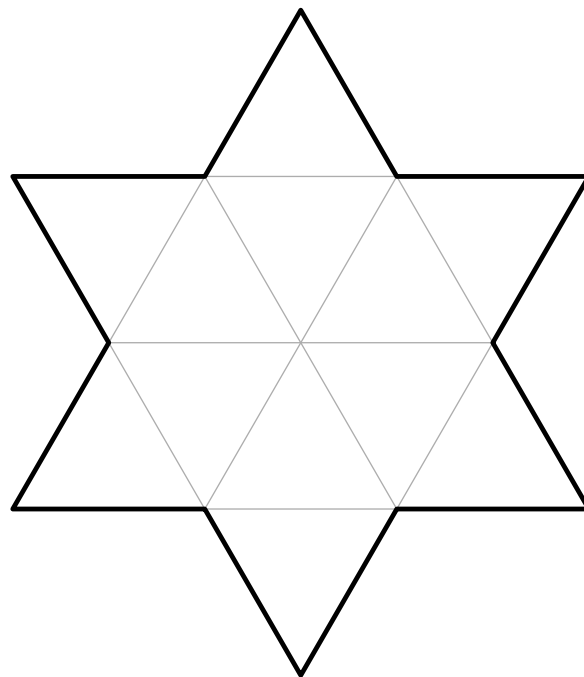
C



Possible Impossible

 _____  _____

D



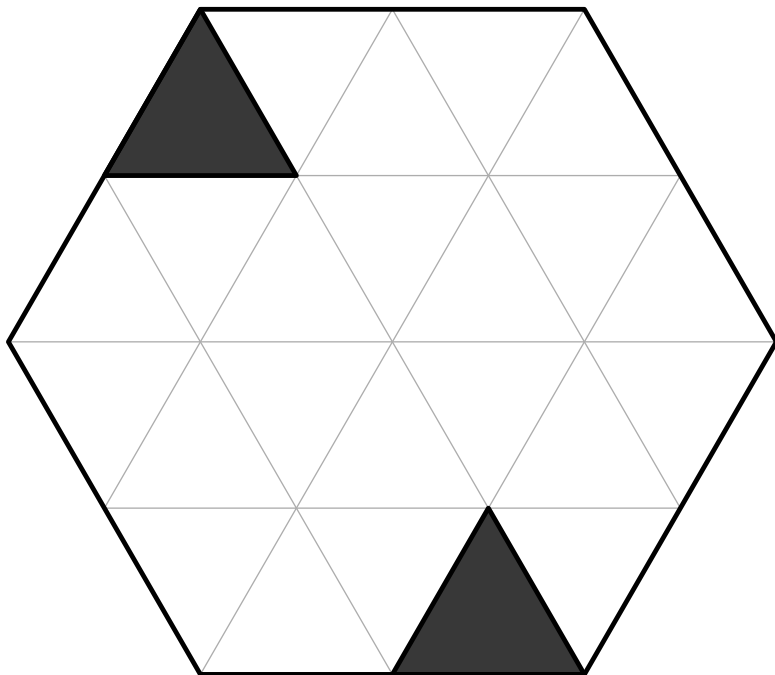
Possible Impossible

 _____  _____

Talk about it. Every board has an even number of triangles. What do the impossible boards have in common? Why does that make them impossible?

3 Broken hexagons

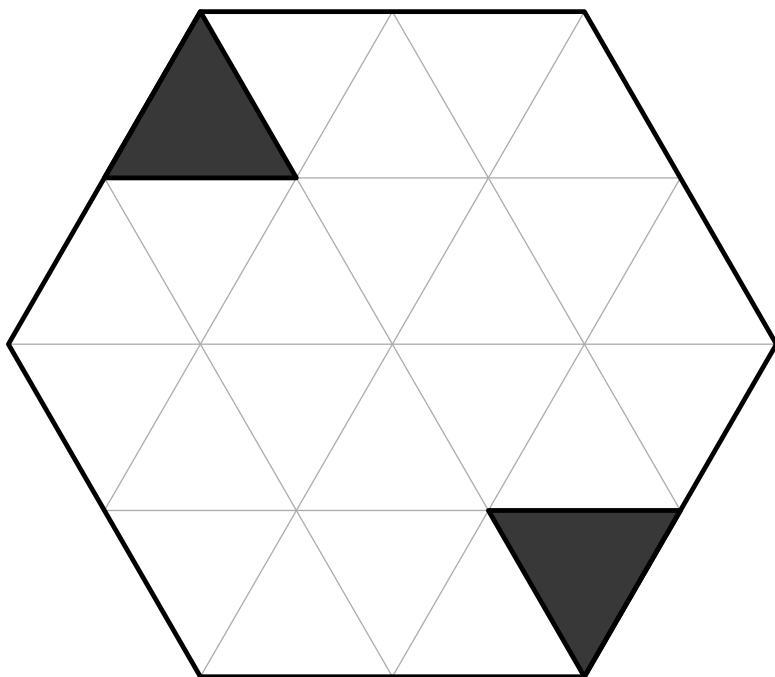
The black triangles are **holes**: no blocks go there. Try to cover all the white triangles with **blue rhombi only** .



X Hexagon X

Possible Impossible

 _____  _____



Y Hexagon Y

Possible Impossible

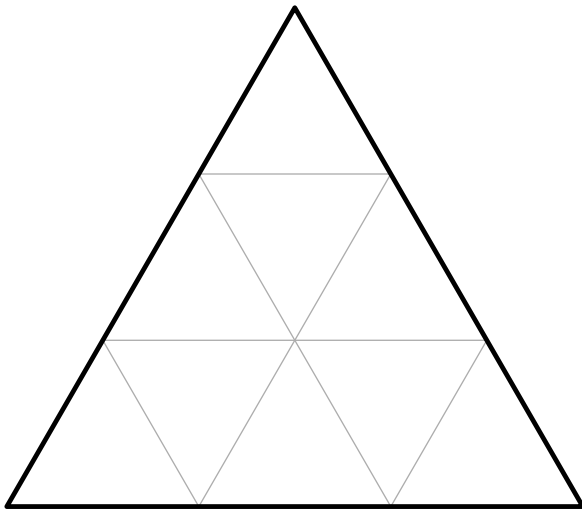
 _____  _____

Both hexagons have 22 white triangles. Why can one of them *never* be covered?

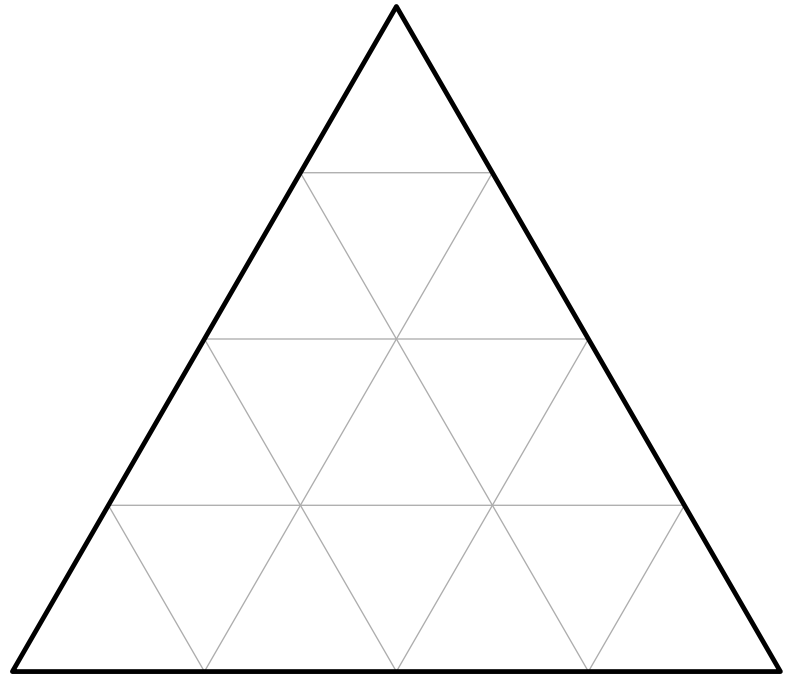
Make your own. Which two triangles could you take out of a whole hexagon so that it is *still* possible? So that it is impossible?

4 Fewest gaps

Can you cover these big triangles with blues alone? If not, put in as **many** blues  as you can, then fill the gaps with greens . Try to use as **few** greens as possible!



Side 3:  _____  _____



Side 4:  _____  _____

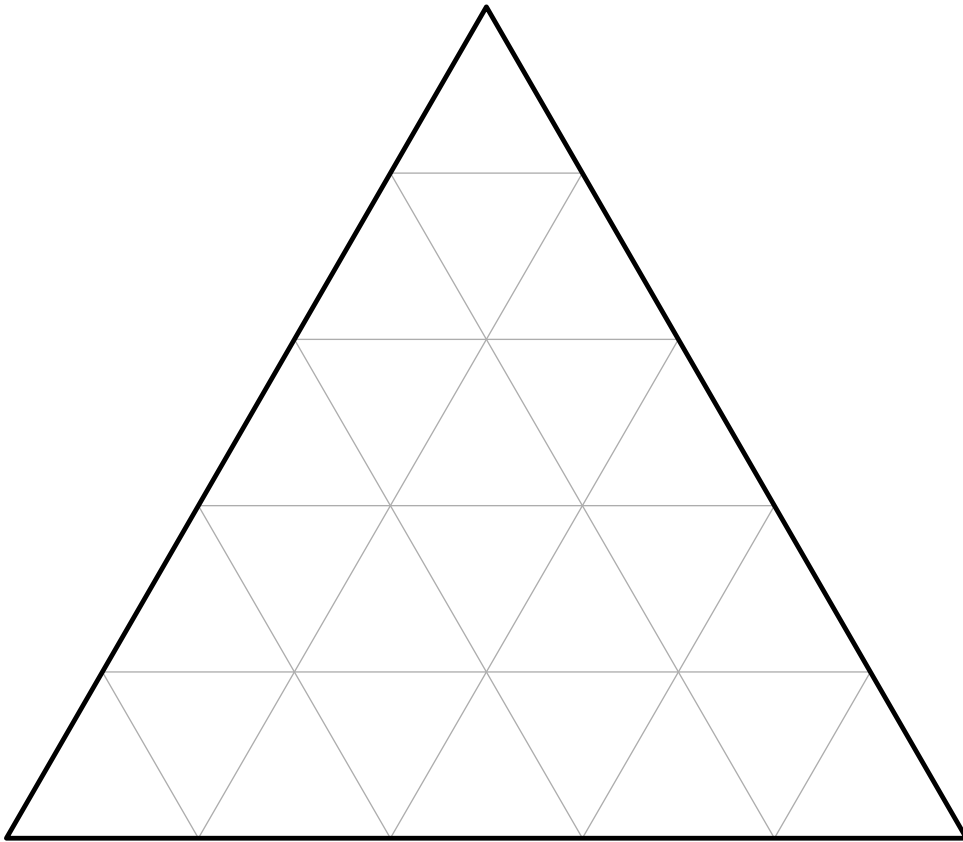
Your best so far. (Board A on page 2 is the side-2 triangle.)

Side	most blues 	fewest greens 
2		
3		
4		

Talk about it. Compare with your group. Can anyone beat your number of greens? Look at the triangles your greens are sitting on. Are they  or .

A plan that always works? Describe a way to place the blues that leaves only as many gaps as the side length. Would your plan work for *every* big triangle?

5 Bigger and bigger



Side 5. Use as many blues as you can. Fill the gaps with greens.



Purple puzzle. Would purple pieces help you leave fewer gaps? Try covering a purple with blues.

Fill in the table. Use your boards for sides 2–5. Can you *predict* sides 6 and 10?

Side	how many 	how many 	most blues 	fewest greens 
2				
3				
4				
5				
6				
10				

Why can nobody ever use fewer greens than this? (Hint: each blue covers one  and one .)
